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Grupa 234

Subiect 66.1

a) Se considera functia compusa $h(t) = g(f(t))$. Sa se exprime conditionarea lui h in functie de conditionarea lui g si f . Atentie la formulare – precizati in care puncte se vor evalua numerele de conditionare.

$$\left. \begin{array}{l} \text{Formula de conditionare : } \text{Cond} = \frac{|x f'(x)|}{|f(x)|} \\ h(t) = g(f(t)) \Rightarrow h'(t) = g'(f(t))f'(t) \end{array} \right\} \Rightarrow \text{Cond}_h = \frac{|th'(t)|}{|h(t)|} \text{ si}$$
$$\text{Cond}_f = \frac{|tf'(t)|}{|f(t)|} \text{ si}$$
$$\text{Cond}_g = \frac{|xg(x)|}{|g(x)|}, \quad x=f(t)$$

$$\text{Cond}_h = \frac{|th'(t)|}{|h(t)|} \Leftrightarrow \text{Cond}_h = \frac{|tg'(f(t))f'(t)|}{|g(f(t))|} \Leftrightarrow \text{Cond}_h = \frac{|f(t)g'(f(t))tf'(t)|}{|f(t)g(f(t))|} \Leftrightarrow$$

$$\text{Cond}_h = \underbrace{\frac{|f(t)g'(f(t))|}{|g(f(t))|}}_{\text{Cond}_g} \underbrace{\frac{|tf'(t)|}{|f(t)|}}_{\text{Cond}_f}$$

Daca :

$$\left. \begin{array}{l} f : \mathbb{R} \rightarrow \mathbb{R}; f(t) = \sin t \\ g : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}; g(t) = \frac{|1+t|}{|1-t|} \end{array} \right\} \Rightarrow \text{Exista } h = g \circ f$$
$$h(t) = g(f(t))$$

b) Ilustrati a) pentru $h(t) = \frac{|1+\sin t|}{|1-\sin t|}, t = \frac{\pi}{4}$

$$\text{Cond}_h = \frac{|th'(t)|}{|h(t)|}$$

$$h'(t) = \frac{|(1+\sin t)'(1-\sin t) - (1-\sin t)'(1+\sin t)|}{|(1-\sin t)^2|} = \frac{|\cos t(1-\sin t) + \cos t(1+\sin t)|}{|(1-\sin t)^2|} = \frac{|2 \cos t|}{|(1-\sin t)^2|}$$

$$h'(t) = \frac{2 \cdot \frac{\sqrt{2}}{2}}{(1 - \frac{\sqrt{2}}{2})^2} = \frac{\sqrt{2}}{\frac{(2-\sqrt{2})^2}{4}} = \frac{4\sqrt{2}}{(2-\sqrt{2})^2} = \frac{4\sqrt{2}}{6-4\sqrt{2}} = \frac{2\sqrt{2}}{3-2\sqrt{2}} = 2\sqrt{2} (3+2\sqrt{2}) = 6\sqrt{2} + 8$$

$$h'\left(\frac{\pi}{4}\right) = 6\sqrt{2} + 8$$

$$h\left(\frac{\pi}{4}\right) = \frac{1+\sin\left(\frac{\pi}{4}\right)}{1-\sin\left(\frac{\pi}{4}\right)} = \frac{1+\frac{\sqrt{2}}{2}}{1-\frac{\sqrt{2}}{2}} = \frac{2+\sqrt{2}}{2-\sqrt{2}} \cdot \frac{2}{2-\sqrt{2}} = \frac{2+\sqrt{2}}{2-\sqrt{2}} = \frac{(2-\sqrt{2})^2}{2} = 3 + 2\sqrt{2}$$

$$\text{Cond}h = \frac{|th'(t)|}{|h(t)|} = \frac{\frac{\pi}{4}(8+6\sqrt{2})}{3+2\sqrt{2}} = \frac{\frac{\pi}{4}(8+6\sqrt{2})(3-2\sqrt{2})}{1} = \frac{\pi}{4} \cdot 2\sqrt{2} = \frac{\pi\sqrt{2}}{2}$$

$$\boxed{\text{Cond}h = \frac{\pi\sqrt{2}}{2}} \quad (1)$$

$$f(t) = \sin t$$

$$f'(t) = \cos t$$

$$f'\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$g(t) = \frac{1+t}{1-t}$$

$$g'(t) = \frac{2}{(1-t)^2}$$

$$g'\left(\frac{\sqrt{2}}{2}\right) = \frac{2}{(1-\frac{\sqrt{2}}{2})^2} = \frac{8}{6-4\sqrt{2}} = 2 \cdot (6+4\sqrt{2})$$

$$g\left(f\left(\frac{\pi}{4}\right)\right) = \frac{2+\sqrt{2}}{2-\sqrt{2}} = \frac{(2+\sqrt{2})^2}{2} = 3 + 2\sqrt{2}$$

$$\left. \begin{aligned} \text{Cond}h &= \frac{|th'(t)|}{|h(t)|} \Rightarrow \text{Cond}h = \frac{\left|\left(\frac{\pi}{4}\right)h'\left(\frac{\pi}{4}\right)\right|}{\left|h\left(\frac{\pi}{4}\right)\right|} \\ \text{Stim ca : } h'(t) &= g'(f(t))f'(t) \end{aligned} \right\} \Rightarrow \text{Cond}h = \frac{\left|\left(\frac{\pi}{4}\right)g'\left(f\left(\frac{\pi}{4}\right)\right)f'\left(\frac{\pi}{4}\right)\right|}{\left|g\left(f\left(\frac{\pi}{4}\right)\right)\right|}$$

$$\text{Cond}h = \frac{\left|\left(\frac{\pi}{4}\right) \cdot g'\left(\frac{\sqrt{2}}{2}\right) \cdot \frac{\sqrt{2}}{2}\right|}{|3+2\sqrt{2}|} = \frac{\frac{\pi}{4} \cdot 2 \cdot (6+4\sqrt{2}) \cdot \frac{\sqrt{2}}{2}}{3+2\sqrt{2}} = \frac{\frac{\pi}{4} \cdot \sqrt{2} \cdot (6+4\sqrt{2})}{3+2\sqrt{2}} = \frac{\pi}{2} (3\sqrt{2} + 4)(3 - 2\sqrt{2})$$

$$\boxed{\text{Cond}h = \frac{\pi\sqrt{2}}{2}} \quad (2)$$

Din (1) si (2) $\Rightarrow$ rezolvarea punctului b)